DS - Set -2

Scenario based

1) * Assign the values of data to a variable called dataset.

* Perform the quantile percentage code for Q_1, Q_2, Q_3 and Q_4.

* Find the IQR by minus the following

$IQR = Q_3 - Q_1$

* Find the outliers by the following code

outlier = dataset [dataset ["columnName"] < (Q_1 - 1.5 x IQR) | (dataset ["columnName"] > (Q_3 + 1.5 * IQR))]

* Print the outliers.

* Use histogram diagram and box plot for a visual representation.

2) * Assign the values of data to a variable called dataset.

* Calculate mean, median, mode for the central tendency.

* After the result, the following should be made if

  * mean > median, it is "Right Skewed"
  * mean < median, it is "Left Skewed"
  * mean = median, the symmetrical.

3) * Assign the values of data to a variable called dataset.

* Divide the sales data into intervals.

* Find how many times it occurs in another column.

* Now visualize using histogram and bar plot.

* Use plt.show() to view the graph output.

4) * Assign the values of data to a variable called dataset.

* Check the dataset. columns

* Use dataset. isnull(). sum to find naN values.

* If found replace it with central tendency (mean, median)

* Now convert all the categorical data to numerical by get_dummies ().

* Check dataset. isnull ( ) .again.

* Now, perform the outliers concept.

* Next, visualize the data using histogram and bar plot.

* Lastly, plt-show ( ) to see the graph.

5) * Assign the values of data to the variable called dataset.

* Assume two parameters H_0 and H_1

* H_0 - Null hypothesis
  H_1 - Alternate hypothesis

* H_0 - There is no difference between medication and placebo.

  H_1 - There is a significant difference between medication and placebo.

* Perform T-Test using the code.
* If p-value is $< 0.05\%$ then, reject null hypothesis.
* If p-value $> 0.05\%$ we fail to reject null hypothesis.

* Lastly conclude the final statement.